

Glutathione reductase activity assessment

Material:

1. Sample
2. Potassium phosphate buffer
3. EDTA
4. NADPH
5. Milli-Q water
6. Oxidized glutathione

Procedures:

1. Homogenize the sample in a sonicator (Branson Sonifier Model 250; three 4-s cycles) in potassium phosphate buffer 100 mM and EDTA 50 mM, pH 7.4, and centrifuge the homogenate at 10.000 rpm and 4 °C for 10 min.
2. Prepare a reaction solution with 2.13 mg of NADPH in 20 mL of potassium phosphate buffer 150 mM (pH 7.0) and EDTA 1.5 mM.
3. Add a 50- μ L aliquot of the sample in a working reagent containing NADPH 0.15 mM, potassium phosphate buffer 0.15 M (pH 7.0), and 100 μ L of water.
4. Analyze the kinetics in a spectrophotometer at 340 nm, for 120 s, at 30-s intervals.
5. Then, add oxidized glutathione, and analysis can continue for 120 s.
6. The enzyme unit was determined as the oxidation of 1 mol of NADPH per minute and was calculated on the basis of the molar absorptivity of NADPH at 340 nm.
7. Normalize the results by the protein content in the sample and expressed as a percentage of the placebo group.